

ZISHUO HUANG

EDUCATION

Department of Electrical Engineering, Tsinghua University

Ph.D. in Electrical Engineering

- Advisor: Prof. Wenchuan Wu
- Research area: optimization and control of renewable energy sources, especially wind farms

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Beijing, China

2022 - 2027 (expected)

School of Electrical Engineering, Xi'an Jiaotong University

B.E. in Electrical Engineering (Qian Xuesen's Class)

Class of the Gifted Young, Xi'an Jiaotong University

Xi'an, China

2018 - 2022

Xi'an, China

2016 - 2018

PUBLICATIONS

1. **Zishuo Huang**, Chenhui Lin, Qi Wang, Wenchuan Wu. A Bilevel Optimization Method for Wind Farm Layout Considering Yaw Control. *IEEE Transactions on Industrial Informatics*, 2025. (IF: 9.9)
2. **Zishuo Huang**, Wenchuan Wu. Dual-Stage MPC-Based AGC for Wind Farm Considering Aerodynamic Interactions. *IEEE Transactions on Sustainable Energy*, 2025. (IF: 10.0)
3. **Zishuo Huang**, Wenchuan Wu, Chenhui Lin, Zizhen Guo. Cooperative Strategies for Frequency Control of Wind Turbines to Mitigate Secondary Frequency Dip: Coefficient Allocation and Exit Techniques. *IEEE Transactions on Sustainable Energy*, 2025. (IF: 10.0)
4. **Zishuo Huang**, Wenchuan Wu. An Efficient Solution for Large Offshore Wind Farm Power Optimization with the Porté-Agel Wake Model: Optimality and Efficiency. *Energy*, 2024. (IF: 9.4)
5. **Zishuo Huang**, Wenchuan Wu, Zizhen Guo. Data-driven Stochastic Model Predictive Control for Regulating Wind Farm Power Generation with Controlled Battery Storage. *IET Renewable Power Generation*, 2023. (IF: 2.9)
6. **Zishuo Huang**, Wenchuan Wu, Hongyan Liu, Jisi Wu, Hao Liu. Real-Time Digital Simulation of DFIG Wind Farm Primary Frequency Control via Droop and Virtual Inertia Control. *6th International Conference on Electrical Technology and Automatic Control (ICETAC)*, 2025.
7. Guojian Li, **Zishuo Huang**, Zhenfeng Wang, Jin Xu, Jihui Dong, Jiang Li. Cooperative Yaw and Induction Control for Wind Farm and Its Effectiveness in Providing Secondary Frequency Regulation. *IEEE 8th Conference on Energy Internet and Energy System Integration (EI2)*, 2024.
8. Guannan Wang, **Zishuo Huang**, Wenchuan Wu, Xiaowei Fan, Jun Zhang, Haitao Liu. Model Predictive Control for Regulating Battery Storage to Smooth the Wind Farm Power Generation. *IEEE International Conference on Power System Technology*, 2023.

PROJECTS

Zhangjiakou 100% Renewable Energy Aggregation and Operational Control: Research & Demonstration

Tsinghua University

2023.04 - 2025.07

In this project, I was responsible for modeling and implementing a plant-level active support control scheme. I first proposed a measurement-driven lifted-mapping approach for global linearization, mapping low-dimensional, highly nonlinear processes to a high-dimensional linear state-space model, which produced an interpretable and tractable dynamic-control prototype and a model-data co-driven framework for power tracking and frequency/voltage support; I then extended the prototype to a **stochastic MPC** with explicit wind-speed forecast-error correction and a deterministic reformulation, unifying in the objective function the penalties on grid-side power fluctuation, schedule-tracking deviation, curtailment, and **battery degradation cost**, while enforcing engineering constraints on **energy storage**, **pitch angle**, and **grid voltage** to realize wind-storage coordinated optimization. The control performance of the cluster controller was validated via real-time hardware-in-the-loop simulation on RTDS.

AWARDS AND HONORS	• National Second Prize (Team Leader) , China Undergraduate Mathematical Contest in Modeling (CUMCM) 2019.11	
	• Provincial First Prize (Shaanxi) , China Undergraduate Mathematical Contest in Modeling (CUMCM) 2018.11	
	• First Prize (Shaanxi Division) , 11th National Undergraduate Mathematics Competition 2019.10	
	• First Prize , Xi'an Jiaotong University Mathematical Modeling Competition	2019.05
	• First Prize , Xi'an Jiaotong University Mathematical Modeling Competition	2018.05
	• Qiu Changrong Scholarship (Second Class) , Academic Year 2019–2020	2020.11
	• Qiu Changrong Scholarship (Second Class) , Academic Year 2018–2019	2019.11
	• Outstanding Student , Qian Xuesen College, Academic Year 2019–2020	2020.10
	• Outstanding Student , Qian Xuesen College, Academic Year 2018–2019	2019.10
		
SKILLS	Languages: Chinese, English.	
	Programming: Python, C++, MATLAB.	
	Simulation softwares: Simulink, RTDS (RSCAD), Powerfactory	
ACADEMIC SERVICES	Reviewers for: <i>IEEE Transactions on Sustainable Energy</i> ,	
	<i>IEEE Transactions on Industrial Informatics</i> ,	
		